

Computer Architecture

Course Code: CSE360

Lecture - 5

Internal Memory

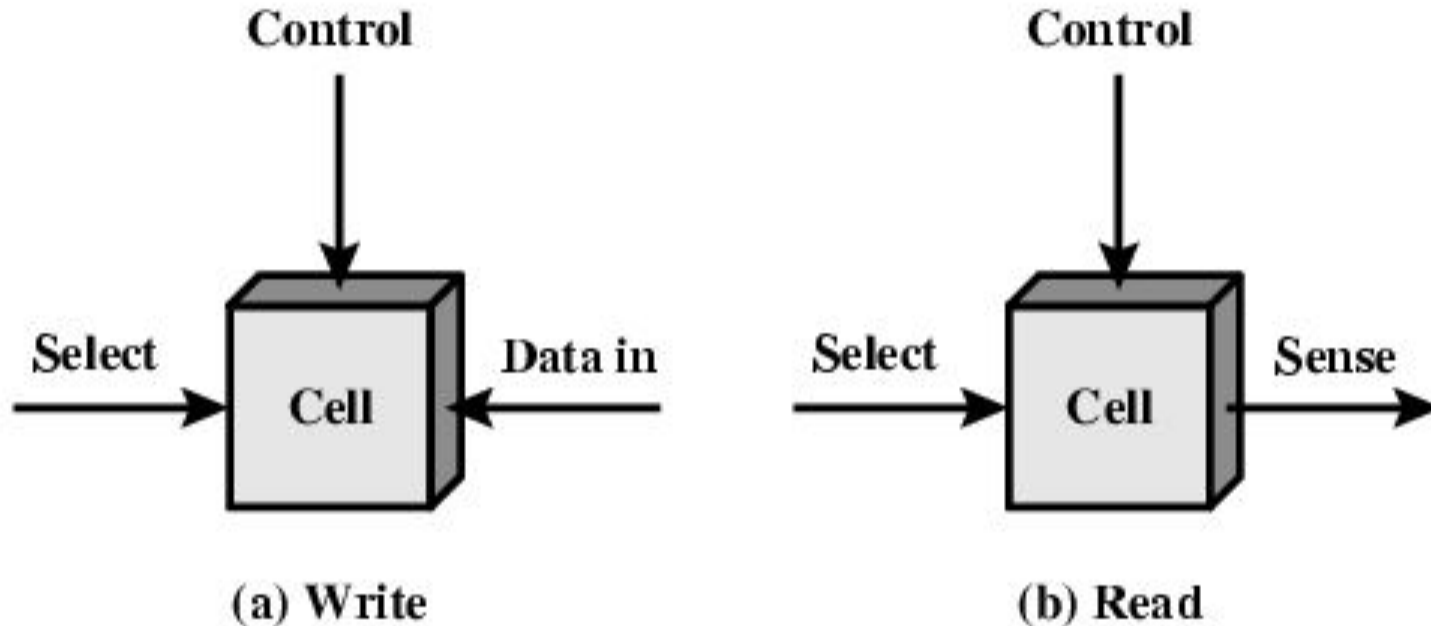
Semiconductor Memory Types

Memory Type	Category	Erasure	Write Mechanism	Volatility
Random-access memory (RAM)	Read-write memory	Electrically, byte-level	Electrically	Volatile
Read-only memory (ROM)	Read-only memory	Not possible	Masks	Nonvolatile
Programmable ROM (PROM)			Electrically	
Erasable PROM (EPROM)	Read-mostly memory	UV light, chip-level		
Electrically Erasable PROM (EEPROM)		Electrically, byte-level		
Flash memory		Electrically, block-level		

Semiconductor Memory

- RAM
 - **Misnamed** as all semiconductor memory is random access
 - **Read data** from the memory and to **write data** into the memory easily and rapidly(through **electrical signals**)
 - **Volatile** (if the power is interrupted, then the data are lost)
 - Can be used as temporary storage
 - Two traditional forms of RAM is Static or Dynamic (**SRAM** or **DRAM**)

Memory Cell Operation



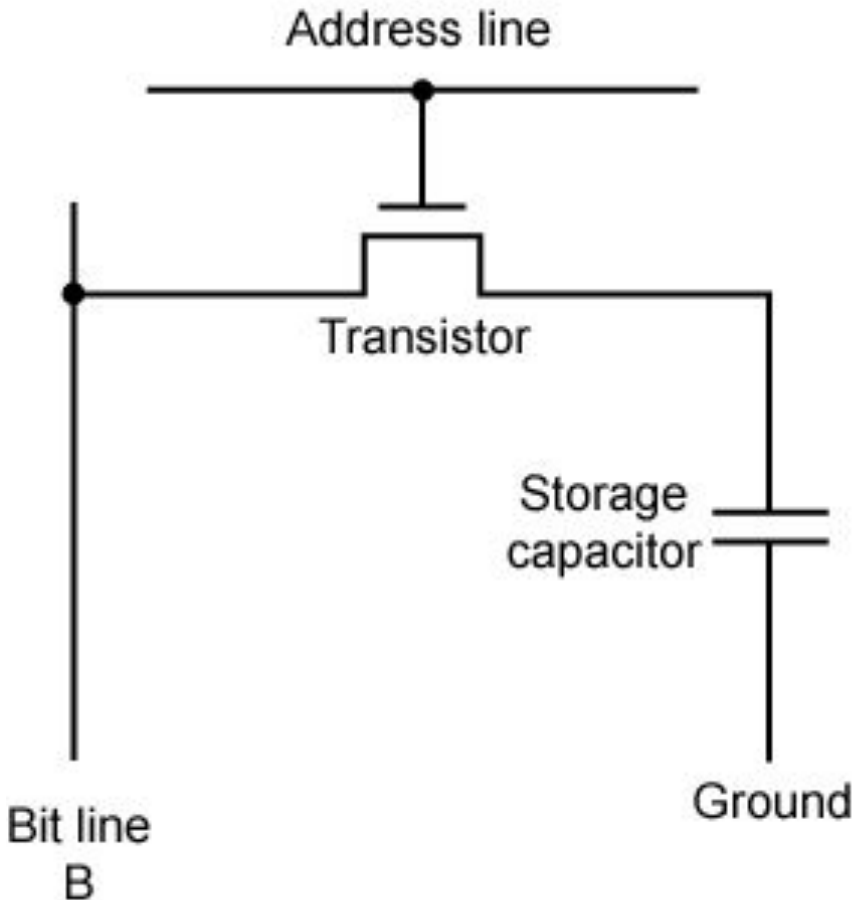
Three terminals carry an electrical signal:

- **Select terminal selects** a memory **cell for read and write** operation
- The **control terminal** indicates **read or write**
- For writing, the other terminal provides an electrical signal that **sets the state of the cell to 1 or 0.**
- For reading, the terminal is used for output of the cell's state.

Dynamic RAM

- **Made with cells that store data as charge on capacitors** (the presence or absence of charge in a capacitor is interpreted **as a binary 1 or 0**)
- **The term dynamic refers to – the tendency of the stored charge to leak away**
- **Capacitors have a natural tendency to discharge; dynamic RAMs need refreshing to maintain data storage**
- **Simpler construction**
- **Smaller per bit**
- **Less expensive**
- **Need refresh circuits**
- **Slower**
- **Main memory**
- The capacitor can store any charge value within a range; **a threshold value determines whether the charge is 0 or 1.**

Dynamic RAM Structure & Operation

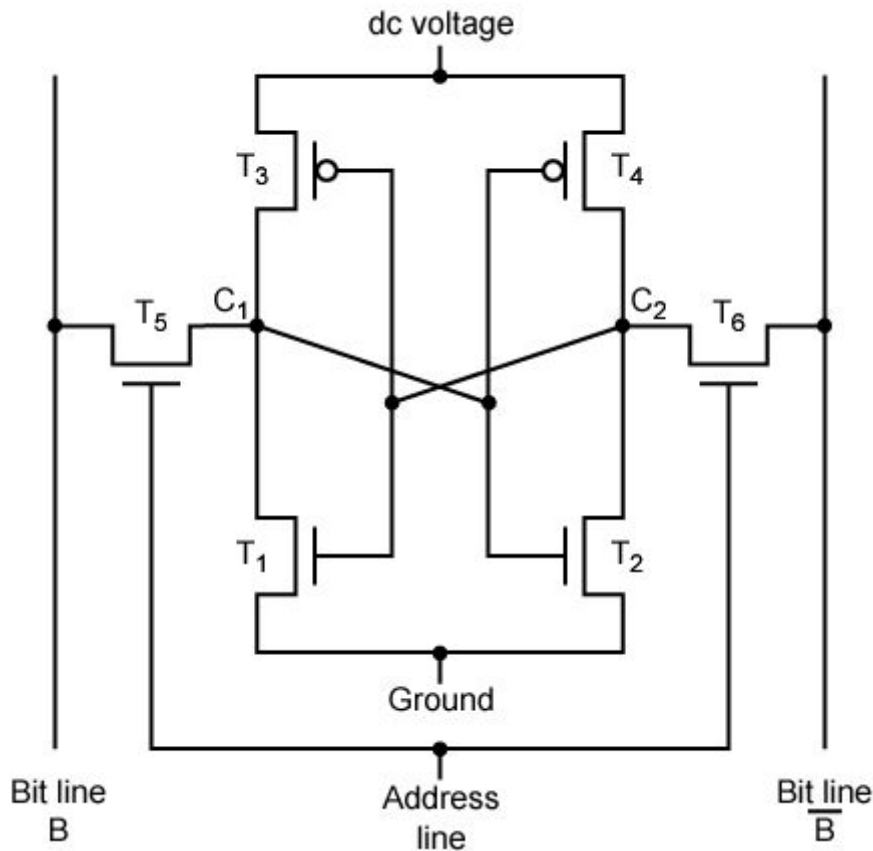


- Figure shows a DRAM structure of an individual **cell that stores one bit**.
- **Address line is activated** when **bit from this cell is to be read or written**
 - Transistor acts a switch that is **closed** (allowing current to flow) if a voltage is applied to the address line and **open** (no current flows) if no voltage is present on the address line
- **Write**
 - A voltage signal applied to bit line
 - High voltage $\square$ 1, low voltage $\square$ 0
 - Then signal is applied to address line
 - Transfers charge to capacitor
- **Read**
 - **Address line is selected**
 - transistor turns on
 - **Charge from capacitor is fed out via bit line and to sense amplifier**
 - **The read out from the cell discharges the capacitor**, which must be **restored** to complete the operation

Static RAM

- **Bits stored as on/off switches**
- **Holds data as long as power is supplied to it**
- **No charges to leak**
- **No refreshing needed when powered**
- **More complex construction**
- **Larger per bit**
- **More expensive**
- **Does not need refresh circuits**
- **Faster**
- **Cache**
- Digital device using the same logic elements in the processor
 - Binary values are stored using **flip-flop logic gate** configuration.

Stating RAM Structure & Operation



- For a SRAM individual cell, **four transistors (T_1, T_2, T_3, T_4) are cross connected.**
- This arrangement gives stable logic state
- **State 1**
 - C_1 high, C_2 low
 - T_1, T_4 off, T_2, T_3 on
- **State 0**
 - C_2 high, C_1 low
 - T_2, T_3 off, T_1, T_4 on
- SRAM **address line is used to open or close a switch.**
- **Address line controls two transistors T_5 and T_6**
- When a signal is applied to this line, the two transistors are switch on, allowing read and write operation.
- **Write – apply the bit value to B & compliment to $\overline{B}$; this forces four transistors (T_1, T_2, T_3, T_4) into proper state.**
- Read – **bit value is read from line B**

SRAM v DRAM

- Both volatile
 - Power needed to preserve data
- **Dynamic cell**
 - **Simpler, smaller** than a static memory cell
 - **More dense** (smaller cells=more cells per unit area)
 - **Less expensive**
 - **Needs refresh circuitry**
 - Tends to be favoured for **large memory** requirements
- **Static**
 - Faster
 - Cache

Read Only Memory (ROM)

- **Permanent storage** (cannot be changed)
 - **Nonvolatile** (no power source is required to maintain the bit values in memory)
 - While **it is possible to read a ROM**, **it is not possible to write** new data into it
- Important Application
 - **Microprogramming (see later)**
 - Library subroutines
 - **Systems programs (BIOS)**
 - Function tables
- Advantage of ROM
 - **Data or program is permanently in main memory and need never be loaded from a secondary storage device.**

Types of ROM

- **Programmable read-only memory (PROM)**
 - Semiconductor memory whose contents may be set only once.
 - The writing process is performed electrically and performed by the user at a time of original chip fabrication
 - Needs special equipment to program
- Read “mostly” – read operations are far more frequent than write operations but for which non-volatile storage is required
 - **Erasable Programmable (EPROM)**
 - Read and write electrically, as with PROM
 - Before writing, all storage cells must be erased by ultraviolet (UV) radiation. [20 minutes take to erase]. After erasing you can write new program or data
 - EPROM is more expensive than PROM but it has advantage of multiple update capability.
 - **Electrically Erasable (EEPROM)**
 - Can be written into anytime without erasing whole contents; only byte or bytes addressed are updated
 - Takes much longer to write than read
 - EEPROM has advantage of non-volatility with flexibility of being updatable
 - More expensive than EPROM and also less dense (supporting fewer bits per cheap)
 - **Flash memory**
 - Erase whole memory or a section (block) electrically within one or a few seconds (in a single action or “**flash**”), which is much faster than EPROM

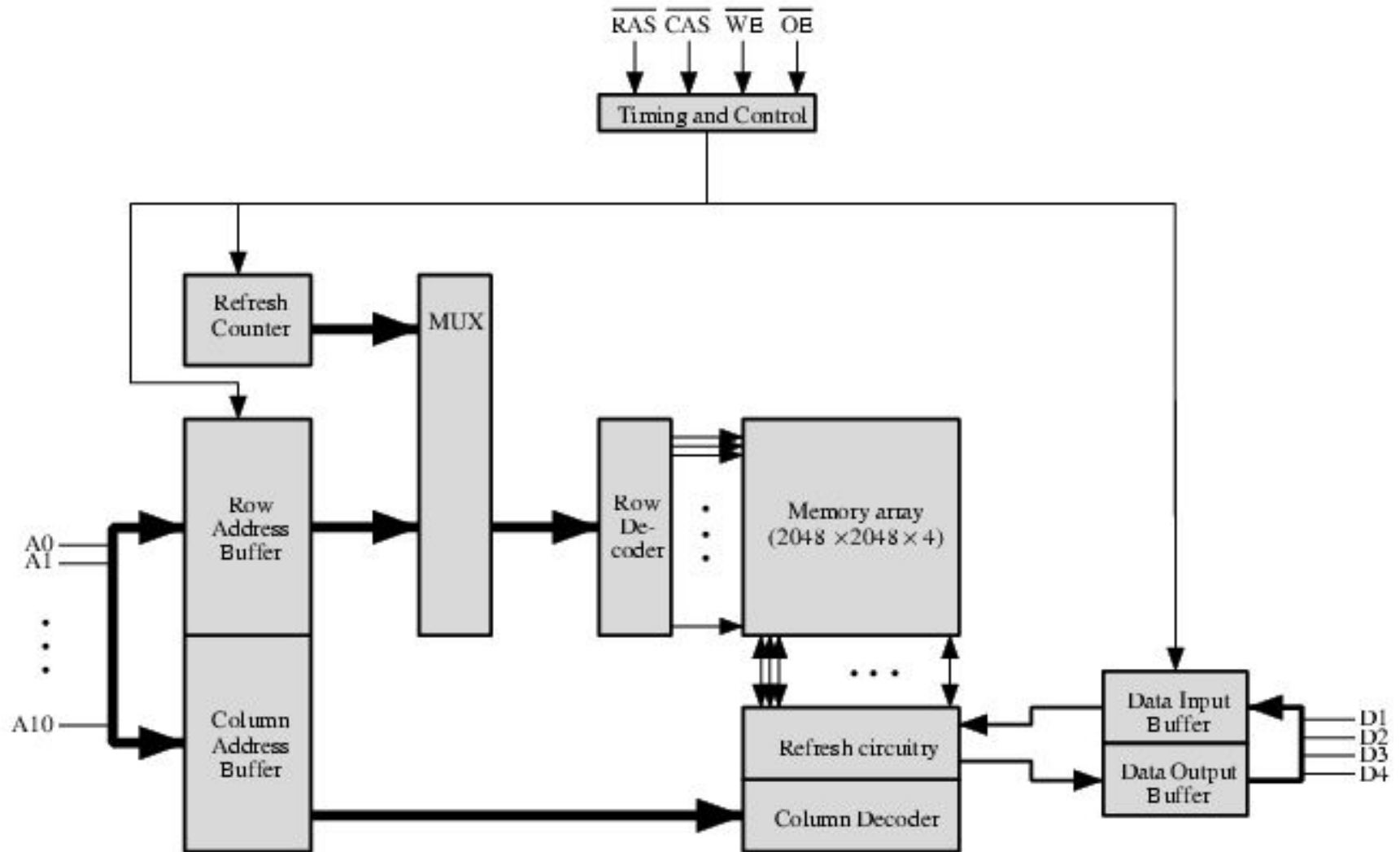
Organisation in detail

- A **16-Mbit (4Mx4)** chip can be organised as **1M of 16 bit** words
- **4 bits** are read and written at a time (**four data lines are used for input and output of 4 bits** to and from a data buffer).
- A **16Mbit** chip can be organised as a **2048 x 2048 x 4bit** array
- **11 address lines are needed to select one of 2048 rows.** These 11 lines are fed into row decoder, which has 11 lines of input and 2048 lines of output. The logic of the **decoder activates one of the 2048 outputs depending on the bit pattern on the 11 input lines ($2^{11}=2048$)**
- **Additional 11 address lines select one of 2048 columns** of 4 bits per columns.
- On input (**write**), **the bit line is activated for a 1 or 0** according to the value of corresponding **data line**.
- On output (**read**), the value of each bit line is passed through a sense amplifier and **presented to the data lines**.
- The **row line selects which row of cells** is used for **reading and writing**
- Reduces number of address pins
- **Multiplex row address and column address** (**first, 11 address signals are passed to define the row address** of the array, then **other 11 are presented for column address**)
- Signals are accompanied by **RAS, CAS, WE, and OE**.
- Adding one more pin doubles the number of rows and columns, so the size of memory grows by a factor of 4

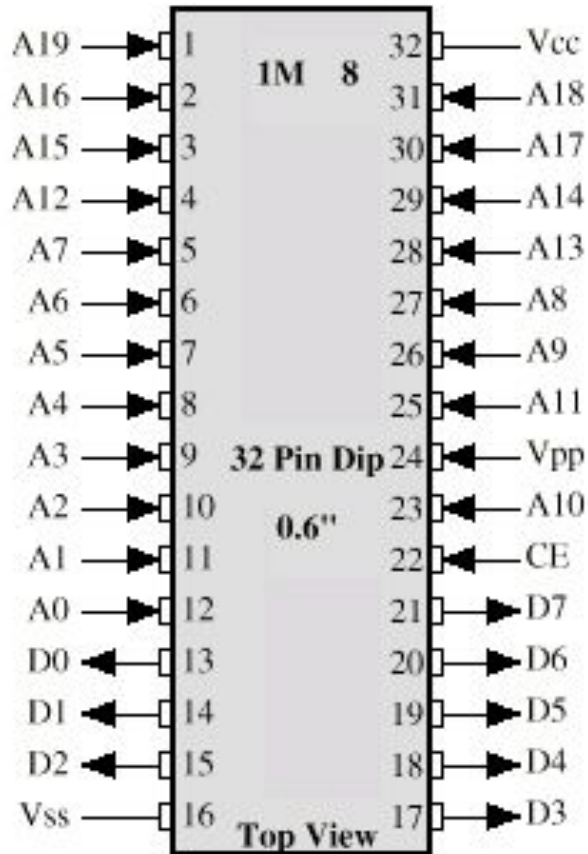
Refreshing

- **All DRAM requires refresh operation**
- For refreshing, to disable the DRAM chip while all data cells are refreshed.
- The refresh counter steps through all row values.
- For each row, output lines from the refresh counter are supplied to the row decoder and RAS line is activated
- **The data are read out and written back into the same location; this causes each cell in the row to be refreshed**

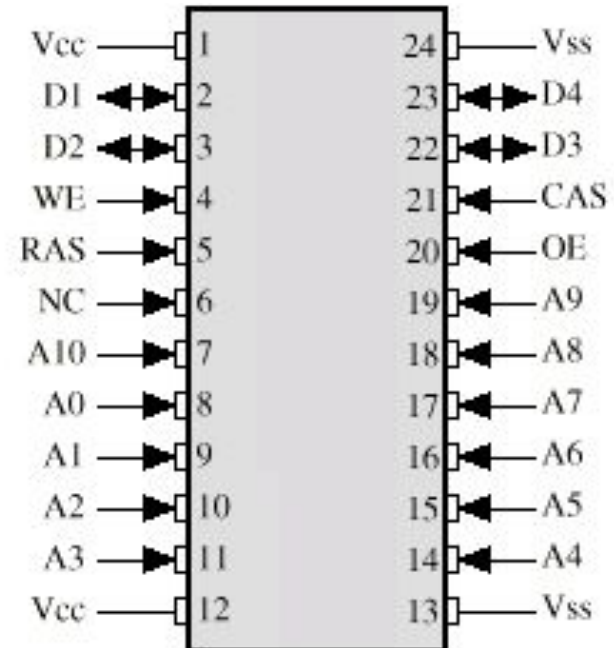
Typical 16 Mb DRAM (4M x 4)



Packaging



(a) 8 Mbit EPROM

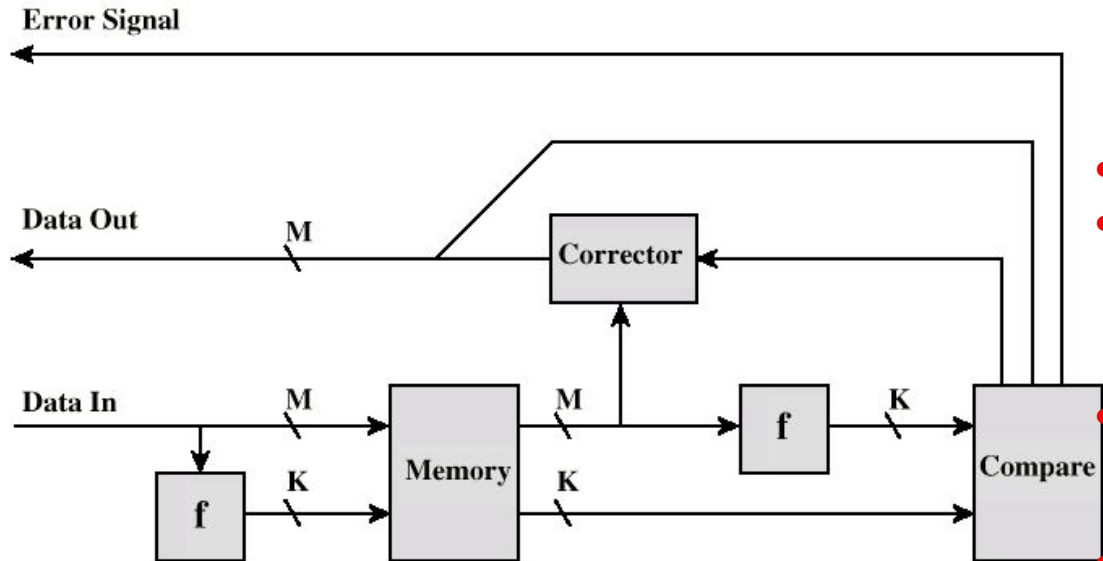


(b) 16 Mbit DRAM

Error Correction

- Hard Failure
 - Permanent defect
- Soft Error
 - Random, non-destructive
 - No permanent damage to memory
- Detected using Hamming error correcting code

Error Correcting Code Function



- When data are to be read into (stored) the memory, a function f is formed on the data to produce a code.
- Both code and data are stored
- Thus, if M -bit word of data is to be stored, and the code is of length K bits; actual size of the stored word is $M+K$ bits
- When stored word is read out, the code is used to detect and possibly correct errors.
- A new set of K code bits is generated from M data bits and compared with the fetched code bits.
- This comparison yields three results:
 - No errors are detected and fetched data bits are sent out.

- An error is detected and possible to correct error. The data bits plus error correction bits are fed into a corrector, which produces M bits to be sent out.
- An error is detected, but it is not possible to correct it. This condition is reported.

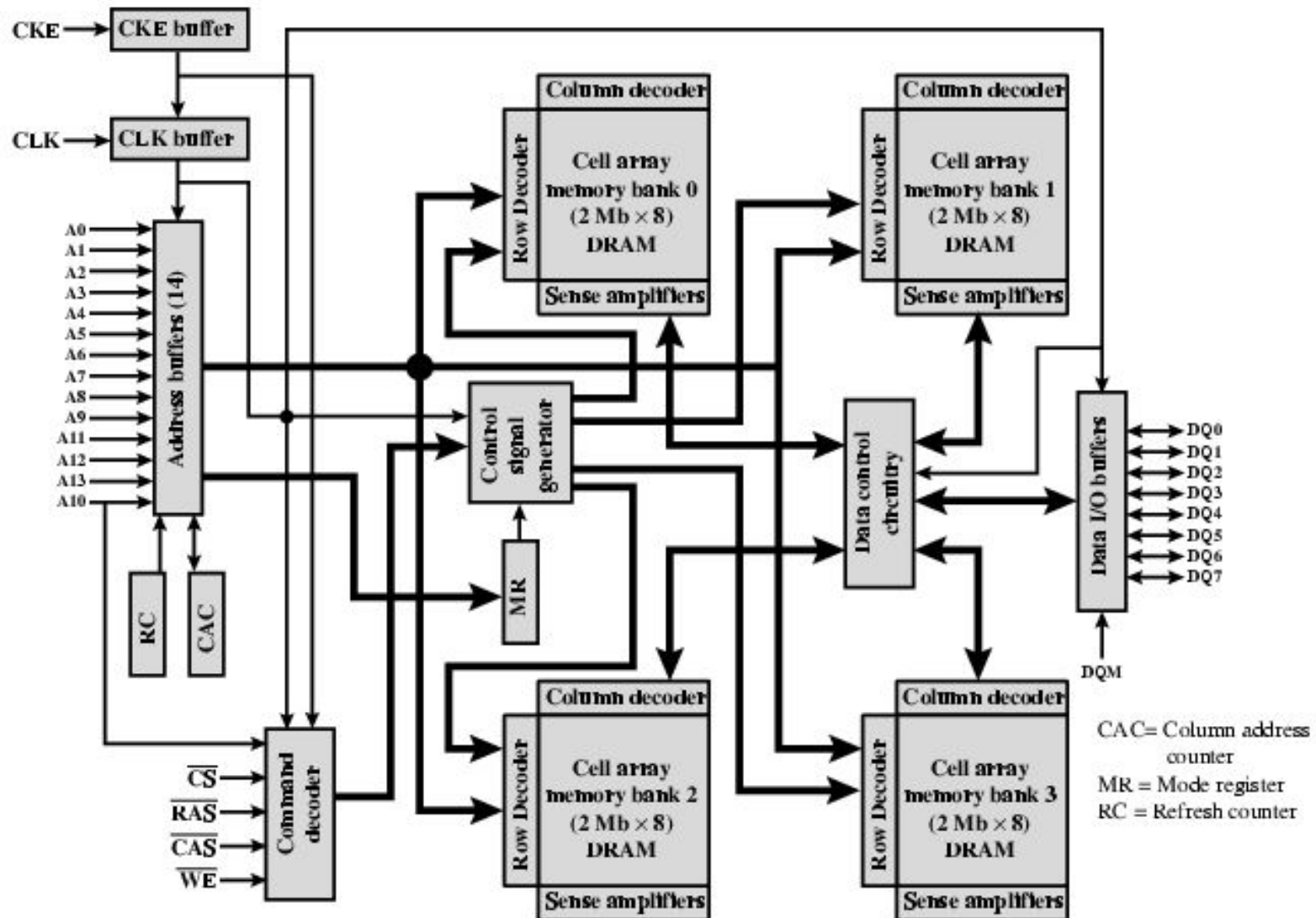
Advanced DRAM Organization

- **Basic DRAM same since first RAM chips**
- **Enhanced DRAM**
 - **Contains small SRAM as well**
 - **SRAM holds last line read (c.f. Cache!)**
- **Cache DRAM**
 - **Larger SRAM component**
 - **Use as cache or serial buffer**

Synchronous DRAM (SDRAM)

- Access is synchronized with an external clock
- Address is presented to RAM
- RAM finds data (CPU waits in conventional DRAM)
- Since SDRAM moves data in time with system clock, CPU knows when data will be ready
- CPU does not have to wait, it can do something else
- In burst mode, a series of data bits can be clocked out rapidly (on chip parallelism) after the first bit has been accessed.
- DDR-SDRAM sends data twice per clock cycle (leading & trailing edge)

SDRAM



DDR SDRAM

- **SDRAM can only send data once per clock**
- **Double-data-rate SDRAM can send data twice per clock cycle**
 - Rising edge and falling edge

Cache DRAM

- Mitsubishi
- Integrates small SRAM cache (16 kb) onto generic DRAM chip
- Used as true cache
 - 64-bit lines
 - Effective for ordinary random access
- To support serial access of block of data
 - E.g. refresh bit-mapped screen
 - CDRAM can prefetch data from DRAM into SRAM buffer
 - Subsequent accesses solely to SRAM